



# **Department of Computer Science & Engineering**

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**Project Report**

**On**

**“Big Data Processing and Analysis of US Accidents Dataset”**

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# **“Big Data Processing and Analysis of US Accidents Dataset”**

## **Abstract**

This project presents a practical implementation of Big Data analytics techniques for processing and analyzing the US Accidents dataset using the Hadoop ecosystem. The dataset contains a large number of real-world accident records collected from across the United States and includes information such as accident severity, start and end time, geographic coordinates, distance, location, weather condition, visibility, traffic-related indicators, and several environmental attributes. Because of its large scale and the variety of attributes available for analysis, the dataset provides a suitable foundation for demonstrating distributed storage and processing technologies in a laboratory environment.

The main purpose of the project is to demonstrate how Hadoop Distributed File System (HDFS), Java-based MapReduce, and Apache Pig can be combined into a complete Big Data processing workflow. HDFS is used as the distributed storage layer where the US Accidents dataset is uploaded and maintained. MapReduce is then used to implement five independent analytical tasks. These tasks calculate state-wise accident counts, severity-wise accident counts, city-wise accident counts, weather-condition frequencies, and traffic-signal-related counts. These analyses demonstrate the lower-level programming model of Hadoop, where Mapper and Reducer logic is explicitly designed to transform and aggregate large numbers of records.

After the MapReduce stage, ten Apache Pig Latin scripts are used to perform additional analyses in a more concise data-flow style. The Pig tasks include filtering severe accidents, calculating state-wise and severity-wise counts, analyzing weather conditions and cities, calculating average accident severity by state, comparing day and night accidents, calculating average accident distance by severity, grouping accidents by timezone, and combining weather condition with severity. Together, these operations demonstrate filtering, grouping, aggregation, projection, ordering, and multi-field analysis on the same HDFS-based dataset. The complete workflow also emphasizes practical Hadoop skills rather than only theoretical concepts. The dataset is first inspected locally, uploaded to HDFS, and verified. Java source files are compiled with the Hadoop libraries and packaged into JAR files before execution. Each MapReduce output is written to a separate HDFS directory, while each Pig script also stores its result in an isolated output location. This organization makes it easier to rerun individual analyses, verify outputs, and document the work through command-line evidence and screenshots.

The project demonstrates that MapReduce and Pig provide complementary approaches to large-scale data analysis. MapReduce offers detailed control over the processing logic through custom Mapper and Reducer classes, while Pig provides a higher-level data-flow language that can express many analytical operations with considerably less code. The resulting workflow shows how a large accident dataset can be transformed from raw CSV records into meaningful statistical summaries that can support understanding of accident frequency, severity, environmental conditions, geographic distribution, and road-related factors. Overall, the project provides a reproducible demonstration of storing, processing, and analyzing large-scale transportation safety data using core Hadoop technologies.

# Contents

|                                                         |    |
|---------------------------------------------------------|----|
| <b>Abstract</b> .....                                   | 1  |
| <b>1. Introduction</b> .....                            | 4  |
| <b>2. Problem Statement</b> .....                       | 4  |
| <b>3. US Accidents Dataset Description</b> .....        | 4  |
| <b>4. Project Objectives</b> .....                      | 5  |
| <b>5. Tools and Technologies</b> .....                  | 6  |
| <b>6. Preparing the Hadoop Environment</b> .....        | 6  |
| 6.1 Implementing HDFS for US Accidents Dataset .....    | 6  |
| 6.2 Create Project Folder .....                         | 7  |
| 6.3 Inspect the CSV .....                               | 7  |
| 6.4 Create HDFS Directories .....                       | 7  |
| 6.5 Upload Dataset to HDFS .....                        | 7  |
| 6.6 Verify Dataset in HDFS .....                        | 8  |
| <b>7. MapReduce Analysis</b> .....                      | 8  |
| 7.1 MapReduce Task 1: State-wise Accident Count .....   | 9  |
| 7.2 MapReduce Task 2: Severity-wise Accident Count..... | 9  |
| 7.3 MapReduce Task 3: City-wise Accident Count .....    | 9  |
| 7.4 MapReduce Task 4: Weather Condition Count.....      | 10 |
| 7.5 MapReduce Task 5: Traffic Signal Count.....         | 10 |
| <b>8. Apache Pig Analysis</b> .....                     | 11 |
| 8.1 Pig Task 1: Severe Accidents .....                  | 11 |
| 8.2 Pig Task 2: State-wise Count.....                   | 12 |
| 8.3 Pig Task 3: Severity-wise Count .....               | 13 |
| 8.4 Pig Task 4: Weather Condition Count.....            | 13 |
| 8.5 Pig Task 5: City-wise Count.....                    | 14 |
| 8.6 Pig Task 6: Average Severity by State .....         | 14 |
| 8.7 Pig Task 7: Day/Night Accident Count.....           | 15 |
| 8.8 Pig Task 8: Average Distance by Severity .....      | 16 |
| 8.9 Pig Task 9: Timezone-wise Count .....               | 16 |
| 8.10 Pig Task 10: Weather Condition and Severity .....  | 17 |
| <b>9. Results and Discussion</b> .....                  | 18 |

|                                              |           |
|----------------------------------------------|-----------|
| <b>10. Limitations and Future Scope.....</b> | <b>19</b> |
| <b>11. Conclusion.....</b>                   | <b>19</b> |
| <b>12. References .....</b>                  | <b>20</b> |

## 1. Introduction

The rapid growth of vehicles, GPS, weather monitoring, and transportation systems generates large amounts of accident data. Analyzing this data can help identify patterns related to accident frequency, severity, location, weather, and road conditions.

This project analyzes the US Accidents dataset using the Hadoop ecosystem. The project uses HDFS for distributed data storage, MapReduce for processing and aggregation, and Apache Pig for high-level data analysis.

Five MapReduce jobs are performed to analyze accidents by state, severity, city, weather condition, and traffic signal. In addition, ten Pig tasks are used for filtering, grouping, counting, averaging, and analyzing different accident characteristics.

Overall, the project demonstrates how Big Data technologies can be used to process a large real-world dataset and extract useful information about road accidents.

## 2. Problem Statement

The US Accidents dataset contains a very large number of accident records and many descriptive attributes. Performing repeated calculations such as counts by state, severity, city, weather condition, and traffic signal using conventional manual methods is inefficient and difficult to maintain. As the dataset grows, the time and memory required for single-machine processing can also increase significantly.

The problem addressed by this project is therefore how to store and process a large accident dataset efficiently while supporting multiple analytical requirements. A scalable environment is required where the raw dataset can be stored reliably and where independent analytical operations can be executed without manually processing every record. Hadoop provides such an environment through HDFS for storage and MapReduce/Pig for distributed processing.

The project specifically aims to solve this problem by placing the dataset into HDFS and applying five MapReduce jobs and ten Pig analyses. The outputs provide summaries that are easier to interpret than the raw CSV and demonstrate the practical use of distributed Big Data processing.

## 3. US Accidents Dataset Description

The US Accidents (2016–2023) dataset is a large-scale traffic accident dataset covering 47 states of the United States. It contains accident-related information such as severity, time, location, distance, weather conditions, and road features. The dataset was collected from February 2016 to March 2023 and contains approximately 7.7 million accident records.

**Dataset Source:** Kaggle

**Dataset Link:** <https://www.kaggle.com/datasets/sobhanmoosavi/us-accidents>

The CSV format makes it suitable for loading into Hadoop and processing with both MapReduce and Apache Pig.

| Field                                                                             | Description                                                   |
|-----------------------------------------------------------------------------------|---------------------------------------------------------------|
| ID                                                                                | Unique identifier of an accident record.                      |
| Source                                                                            | Source from which the accident record was obtained.           |
| Severity                                                                          | Accident severity category.                                   |
| Start_Time / End_Time                                                             | Beginning and ending timestamps associated with the accident. |
| Start_Lat / Start_Lng                                                             | Latitude and longitude of the accident start location.        |
| End_Lat / End_Lng                                                                 | Latitude and longitude of the accident end location.          |
| Distance                                                                          | Distance associated with the accident record.                 |
| Description                                                                       | Textual description of the accident.                          |
| Street / City / County / State                                                    | Geographic location information.                              |
| Weather_Condition                                                                 | Recorded weather condition.                                   |
| Temperature / Humidity / Pressure / Visibility                                    | Environmental measurements.                                   |
| Wind_Direction / Wind_Speed / Precipitation                                       | Wind and precipitation information.                           |
| Amenity, Bump, Crossing, Give_Way, Junction, No_Exit                              | Road/environment feature indicators.                          |
| Railway, Roundabout, Station, Stop, Traffic_Calming, Traffic_Signal, Turning_Loop | Additional road and traffic feature indicators.               |
| Sunrise_Sunset / Civil_Twilight / Nautical_Twilight / Astronomical_Twilight       | Time-of-day and twilight indicators.                          |

The project uses the complete 47-field schema in Pig, with appropriate numeric and character data types. The schema is important because it determines which fields can be grouped, counted, filtered, or averaged during analysis.

## 4. Project Objectives

- Store the large US Accidents CSV dataset in HDFS and verify that it is accessible.
- Demonstrate Java-based Hadoop MapReduce processing through five independent analytical jobs.
- Use Apache Pig to perform ten higher-level data-flow analyses on the same HDFS dataset.
- Generate separate, verifiable HDFS outputs for every analysis.
- Compare the suitability of low-level MapReduce logic with concise Pig Latin operations.
- Extract meaningful patterns related to accident location, severity, weather, distance, time of day, and road features.
- Document a complete and reproducible Big Data workflow suitable for laboratory evaluation.

## 5. Tools and Technologies

- **Hadoop 3.2.4** — distributed storage and processing framework.
- **Hadoop Distributed File System (HDFS)** — storage layer for the dataset and outputs.
- **Java/JDK** — implementation language for MapReduce programs.
- **Apache Pig 0.17.0** — high-level data-flow analysis using Pig Latin.
- **Windows Command Prompt** — project execution and administration environment.
- **CSV** — input data format.

The environment was configured so that Hadoop commands, Java compilation, HDFS commands, MapReduce jobs, and Pig scripts could be executed from Command Prompt. During execution, the Windows Hadoop environment required the session variable `HADOOP_USER_NAME=TechSource` to avoid a username/path-space issue in YARN container execution.

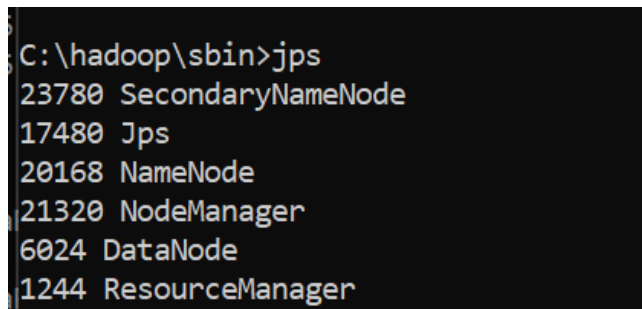
## 6. Preparing the Hadoop Environment

Before processing the dataset, the Hadoop environment was checked and the required services were started. Java was required for Hadoop and MapReduce compilation, Hadoop was required for HDFS and YARN, and Apache Pig was required for the ten Pig analyses. Version checks were performed from Command Prompt. Hadoop was confirmed as version 3.2.4 and Pig as version 0.17.0. The main local project directory was organized as `C:\BigDataProject` with source files under `C:\BigDataProject\src`, compiled classes under `C:\BigDataProject\classes`, and dataset files under `C:\BigDataProject\dataset`.

### 6.1 Implementing HDFS for US Accidents Dataset

#### Start Hadoop

Hadoop services were started so that HDFS and YARN were available. The running Java processes were checked with `jps`. The expected services include NameNode, DataNode, ResourceManager, and NodeManager.



```
C:\hadoop\sbin>jps
23780 SecondaryNameNode
17480 Jps
20168 NameNode
21320 NodeManager
6024 DataNode
1244 ResourceManager
```

## 6.2 Create Project Folder

The local project was maintained under C:\BigDataProject. Separate folders were used for MapReduce, Pig scripts, dataset files, and output-related files to keep the workflow organized.

```
C:\hadoop\sbin>cd /d C:\BigDataProject
C:\BigDataProject>tree /F
Folder PATH listing for volume windows10
Volume serial number is 963B-8F97
C:.
  .gitignore
  readme.html
  --
classes
  CityCount$CityMapper.class
  CityCount$CityReducer.class
  CityCount.class
  SeverityCount$SeverityMapper.class
  SeverityCount$SeverityReducer.class
  SeverityCount.class
  StateCount$StateMapper.class
  StateCount$StateReducer.class
  StateCount.class
  TrafficSignalCount$TrafficSignalMapper.class
  TrafficSignalCount$TrafficSignalReducer.class
  TrafficSignalCount.class
  WeatherCount$WeatherMapper.class
  WeatherCount$WeatherReducer.class
  WeatherCount.class
  --
dataset
  readme.html
```

## 6.3 Inspect the CSV

The US Accidents CSV files were inspected before HDFS upload. The header was checked to confirm the expected fields and comma-separated structure. Inspecting the raw input before processing is important because malformed headers, missing fields, or damaged files can cause downstream MapReduce and Pig jobs to fail.

```
C:\BigDataProject\dataset>dir
Volume in drive C is Windows10
Volume Serial Number is 963B-8F97

Directory of C:\BigDataProject\dataset

09/06/2026  11:19 PM    <DIR>          .
09/06/2026  10:22 AM    <DIR>          ..
09/06/2026  11:40 PM             12,152  readme.html
09/06/2026  11:18 PM      566,994,377  US_Accidents_1.5M.csv
09/06/2026  11:09 PM    3,058,183,727  US_Accidents_March23.csv
               3 File(s)  3,625,190,256 bytes
               2 Dir(s)  42,057,560,064 bytes free
```

## 6.4 Create HDFS Directories

The main HDFS workspace was created under /usaccidents. The input directory was created at /usaccidents/input and separate output directories were used for MapReduce and Pig results.

Example commands:

```
set HADOOP_USER_NAME=TechSource
```

```
hdfs dfs -mkdir -p /usaccidents/input
```

```
hdfs dfs -mkdir -p /usaccidents/output_state
```

```
C:\BigDataProject\dataset>hdfs dfs -mkdir -p /usaccidents/input
C:\BigDataProject\dataset>hdfs dfs -ls /usaccidents
Found 1 items
drwxr-xr-x  - Tech Source supergroup          0 2026-09-06 23:27 /usaccidents/input
C:\BigDataProject\dataset>hdfs dfs -put US_Accidents_1.5M.csv /usaccidents/input/
C:\BigDataProject\dataset>hdfs dfs -ls -h /usaccidents/input
Found 1 items
-rw-r--r--  1 Tech Source supergroup    548.7 M 2026-09-06 23:28 /usaccidents/input/US_Accidents_1.5M.csv
C:\BigDataProject\dataset>hdfs dfs -ls -h /usaccidents/input
Found 1 items
-rw-r--r--  1 Tech Source supergroup    548.7 M 2026-09-06 23:28 /usaccidents/input/US_Accidents_1.5M.csv
C:\BigDataProject\dataset>powershell -NoProfile -Command "Get-Content 'US_Accidents_1.5M.csv' -TotalCount 2"
ID,Source,Severity,Start_Time,End_Time,Start_Lat,Start_Lng,End_Lat,End_Lng,Distance(mi),Description,Street,City,County,S
tate,Zipcode,Country,Timezone,Airport_Code,Weather_Timestamp,Temperature(F),Wind_Chill(F),Humidity(%),Pressure(in),Visib
ility(mi),Wind_Direction,Wind_Speed(mph),Precipitation(in),Weather_Condition,Amenity,Bump,Crossing,Give_Way,Junction,No
Exit,Railway,Roundabout,Station,Stop,Traffic_Calming,Traffic_Signal,Turning_Loop,Sunrise,Sunset,Civil_Twilight,Nautical
Twilight,Astronomical_Twilight
A-1 Source2 3 2016-02-08 05:46:00 2016-02-08 11:00:00 39.855147 -84.858723 -84.858723 0 01 Right lane blocked due to accident on
```

## 6.5 Upload Dataset to HDFS

The US Accidents CSV dataset was uploaded from the local dataset folder into /usaccidents/input using hdfs dfs -put or hdfs dfs -copyFromLocal. After upload, the dataset

became available to Hadoop jobs through the HDFS path rather than only through the local Windows filesystem.

```
C:\Windows\system32>hdfs dfs -chmod 777 /usaccidents  
  
C:\Windows\system32>hdfs dfs -ls /usaccidents  
Found 1 items  
drwxr-xr-x  - Tech Source supergroup      0 2026-09-06 23:28 /usaccidents/input
```

## 6.6 Verify Dataset in HDFS

The uploaded file was verified using HDFS listing and file inspection commands. This verification step ensures that the input exists at the expected path before running the analytical jobs. The HDFS input directory was reused by both MapReduce and Pig analyses.

```
C:\Windows\system32>hdfs dfs -ls /usaccidents  
Found 1 items  
drwxr-xr-x  - Tech Source supergroup      0 2026-09-06 23:28 /usaccidents/input
```

## 7. MapReduce Analysis

Five Java MapReduce programs were prepared for the project. Each program follows the Mapper/Reducer processing model and is packaged into a JAR before execution. The jobs read the common HDFS input and write their results to separate HDFS output directories. The Java source files include StateCount.java, SeverityCount.java, CityCount.java, WeatherCount.java, and TrafficSignalCount.java.

The Hadoop compilation classpath included the common, HDFS, MapReduce, and YARN libraries. After compilation, the class files were packaged into JAR files such as StateCount.jar, SeverityCount.jar, CityCount.jar, WeatherCount.jar, and TrafficSignalCount.jar. The use of separate JARs keeps each analytical task simple and independently executable

1. State-wise Accident Count
2. Severity-wise Accident Count
3. City-wise Accident Count
4. Weather Condition Analysis
5. Traffic Signal-wise Accident Count

## Browse Directory

/usaccidents

Go!

Show

25

entries

Search:

| <input type="checkbox"/> |  | Permission | Owner       | Group      | Size | Last Modified | Replication | Block Size | Name            |  |
|--------------------------|--|------------|-------------|------------|------|---------------|-------------|------------|-----------------|--|
| <input type="checkbox"/> |  | drwxr-xr-x | Tech Source | supergroup | 0 B  | Sep 06 23:28  | 0           | 0 B        | input           |  |
| <input type="checkbox"/> |  | drwxr-xr-x | TechSource  | supergroup | 0 B  | Sep 07 11:15  | 0           | 0 B        | output_city     |  |
| <input type="checkbox"/> |  | drwxr-xr-x | TechSource  | supergroup | 0 B  | Sep 07 11:04  | 0           | 0 B        | output_severity |  |
| <input type="checkbox"/> |  | drwxr-xr-x | TechSource  | supergroup | 0 B  | Sep 07 11:38  | 0           | 0 B        | output_signal   |  |
| <input type="checkbox"/> |  | drwxr-xr-x | TechSource  | supergroup | 0 B  | Sep 06 23:54  | 0           | 0 B        | output_state    |  |
| <input type="checkbox"/> |  | drwxr-xr-x | TechSource  | supergroup | 0 B  | Sep 07 11:31  | 0           | 0 B        | output_weather  |  |

### 7.1 MapReduce Task 1: State-wise Accident Count

The StateCount program reads the State field and emits the state as the key with a count of one. The Reducer sums the values for each state. The output is written to /usaccidents/output\_state. This analysis provides a state-level view of accident frequency.

```
C:\Windows\system32>hdfs dfs -ls /usaccidents/output_state
Found 2 items
-rw-r--r-- 1 TechSource supergroup 0 2026-09-06 23:54 /usaccidents/output_state/_SUCCESS
-rw-r--r-- 1 TechSource supergroup 405 2026-09-06 23:54 /usaccidents/output_state/part-r-00000

C:\Windows\system32>hdfs dfs -cat /usaccidents/output_state/part-r-00000
AL 33087
AR 984
AZ 22921
CA 336555
CO 13343
CT 11275
DC 1873
DE 2718
FL 142758
GA 45434
IA 5582
ID 70
IL 60311
IN 8304
KS 2344
```

### 7.2 MapReduce Task 2: Severity-wise Accident Count

The SeverityCount program extracts the Severity field and counts how many records belong to each severity category. This gives a compact distribution of accident seriousness across the dataset.

```
C:\BigDataProject\src>notepad SeverityCount.java

C:\BigDataProject\src>javac -cp "C:\hadoop\share\hadoop\common\*;C:\hadoop\share\hadoop\common\lib\*;C:\hadoop\share\hadoop\hdfs\lib\*;C:\hadoop\share\hadoop\mapreduce\*;C:\hadoop\share\hadoop\mapreduce\lib\*;C:\hadoop\share\hadoop\yarn\*;C:\hadoop\share\hadoop\yarn\lib\*" -d ..\classes SeverityCount.java

C:\BigDataProject\src>dir ..\classes\SeverityCount*
Volume in drive C is Windows10
Volume Serial Number is 9638-8F97

Directory of C:\BigDataProject\classes

09/07/2026 11:00 AM 1,843 SeverityCount$SeverityMapper.class
09/07/2026 11:00 AM 1,755 SeverityCount$SeverityReducer.class
09/07/2026 11:00 AM 1,474 SeverityCount.class
                3 File(s) 5,072 bytes
                0 Dir(s) 42,453,807,394 bytes free

C:\BigDataProject\src>jar -cvf ../SeverityCount.jar -C ..\classes .
added manifest
adding: SeverityCount$SeverityMapper.class(in = 1843) (out= 791)(deflated 57%)
adding: SeverityCount$SeverityReducer.class(in = 1755) (out= 742)(deflated 57%)
adding: SeverityCount.class(in = 1474) (out= 768)(deflated 48%)
adding: StateCount$StateMapper.class(in = 1917) (out= 845)(deflated 55%)
adding: StateCount$StateReducer.class(in = 1748) (out= 737)(deflated 57%)
adding: StateCount.class(in = 1627) (out= 902)(deflated 44%)
```

### 7.3 MapReduce Task 3: City-wise Accident Count

The CityCount program extracts City and counts accident records for each city. Because city names occur across many records, the Reducer aggregates identical city keys into total counts.

```
C:\BigDataProject\src>javac -cp "C:\hadoop\share\hadoop\common\*;C:\hadoop\share\hadoop\common\lib\*;C:\hadoop\share\hadoop\hdfs\*;C:\hadoop\share\hadoop\mapreduce\*;C:\hadoop\share\hadoop\yarn\*;C:\hadoop\share\hadoop\lib\*" -d ..\classes CityCount.java
javac: file not found: CityCount.java
Usage: javac options<source files>
Use -help for a list of possible options

C:\BigDataProject\src>dir
Volume in drive C is Windows10
Volume Serial Number is 9638-8f97

Directory of C:\BigDataProject\src

09/07/2026  11:10 AM    <DIR>          .
09/07/2026  11:01 AM    <DIR>          ..
09/07/2026  11:10 AM                2,532 CityCount.txt
09/07/2026  11:00 AM                2,388 SeverityCount.java
09/06/2026  11:38 PM                2,826 StateCount.java
               3 File(s)              7,746 bytes free
               2 Dir(s)  42,468,601,856 bytes free

C:\BigDataProject\src>ren CityCount.txt CityCount.java

C:\BigDataProject\src>javac -cp "C:\hadoop\share\hadoop\common\*;C:\hadoop\share\hadoop\common\lib\*;C:\hadoop\share\hadoop\hdfs\*;C:\hadoop\share\hadoop\mapreduce\*;C:\hadoop\share\hadoop\yarn\*;C:\hadoop\share\hadoop\lib\*" -d ..\classes CityCount.java
```

## 7.4 MapReduce Task 4: Weather Condition Count

The WeatherCount program extracts Weather\_Condition and counts records for each non-empty weather category. This helps summarize the environmental conditions appearing in the accident data.

```
C:\BigDataProject\src>jar -cvf ../WeatherCount.jar -C ../classes WeatherCount.class -C ../classes WeatherCount$WeatherMapper.class -C ../classes WeatherCount$WeatherReducer.class
added manifest
adding: WeatherCount.class(in = 1475) (out= 796)(deflated 46%)
adding: WeatherCount$WeatherMapper.class(in = 1885) (out= 835)(deflated 55%)
adding: WeatherCount$WeatherReducer.class(in = 1750) (out= 739)(deflated 57%)

C:\BigDataProject\src>hadoop jar ../WeatherCount.jar WeatherCount /usaccidents/input /usaccidents/output_weather
2026-09-07 11:29:27,803 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0:8032
2026-09-07 11:29:29,842 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-09-07 11:29:28,863 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/tech-source/staging/job_1788716948664_0007
2026-09-07 11:29:29,175 INFO input.FileInputFormat: Total input files to process : 1
2026-09-07 11:29:29,266 INFO mapreduce.JobSubmitter: number of splits:5
2026-09-07 11:29:29,450 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1788716948664_0007
2026-09-07 11:29:29,453 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-09-07 11:29:29,730 INFO conf.Configuration: resource-types.xml not found
2026-09-07 11:29:29,731 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-09-07 11:29:29,841 INFO impl.YarnClientImpl: Submitted application application_1788716948664_0007
2026-09-07 11:29:29,886 INFO mapreduce.Job: The url to track the job: http://NASRIN:8088/proxy/application_1788716948664_0007/
2026-09-07 11:29:29,887 INFO mapreduce.Job: Running job: job_1788716948664_0007
2026-09-07 11:29:34,973 INFO mapreduce.Job: Job job_1788716948664_0007 running in user mode : false
```

## 7.5 MapReduce Task 5: Traffic Signal Count

The TrafficSignalCount program uses the Traffic\_Signal field and counts available category values. The program skips the CSV header and ignores empty values so that the aggregation represents actual records rather than the column name.

```
C:\BigDataProject\src>notepad TrafficSignalCount.java

C:\BigDataProject\src>javac -cp "C:\hadoop\share\hadoop\common\*;C:\hadoop\share\hadoop\common\lib\*;C:\hadoop\share\hadoop\hdfs\*;C:\hadoop\share\hadoop\mapreduce\*;C:\hadoop\share\hadoop\yarn\*;C:\hadoop\share\hadoop\lib\*" -d ..\classes TrafficSignalCount.java

C:\BigDataProject\src>jar -cvf ../TrafficSignalCount.jar -C ../classes TrafficSignalCount.class -C ../classes TrafficSignalCount$TrafficSignalMapper.class -C ../classes TrafficSignalCount$TrafficSignalReducer.class
added manifest
adding: TrafficSignalCount.class(in = 1520) (out= 796)(deflated 47%)
adding: TrafficSignalCount$TrafficSignalMapper.class(in = 1914) (out= 831)(deflated 56%)
adding: TrafficSignalCount$TrafficSignalReducer.class(in = 1780) (out= 745)(deflated 58%)

C:\BigDataProject\src>hadoop jar ../TrafficSignalCount.jar TrafficSignalCount /usaccidents/input /usaccidents/output_traffic_signal
2026-09-07 11:36:15,983 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0:8032
2026-09-07 11:36:16,945 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-09-07 11:36:16,966 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/tech-source/staging/job_1788716948664_0009
2026-09-07 11:36:17,236 INFO input.FileInputFormat: Total input files to process : 1
2026-09-07 11:36:17,328 INFO mapreduce.JobSubmitter: number of splits:5
```

Verified MapReduce Example: State-wise Output

The verified StateCount output contains the following selected state totals. These values are included as actual project output rather than estimated figures.

| State | Accident Count |
|-------|----------------|
| CA    | 336,555        |
| TX    | 164,364        |
| FL    | 142,758        |
| SC    | 83,286         |

|    |        |
|----|--------|
| PA | 64,569 |
| NY | 62,698 |
| IL | 60,311 |
| NC | 51,521 |
| GA | 45,434 |
| MI | 46,102 |

The complete verified output also includes all other states present in the dataset. California has the highest count among the listed results, followed by Texas and Florida, indicating a strong concentration of recorded accidents in these states.

## 8. Apache Pig Analysis

Apache Pig was used to perform ten additional analyses on the US Accidents data. Pig Latin provides a concise way to load the CSV with a defined schema, filter records, group records by one or more fields, calculate aggregates, and store results back to HDFS. Each task uses its own HDFS output path so that the results remain isolated and can be verified independently.

Ten pig analyses were performed:

1. Severe Accident Filtering
2. State-wise Accident Count
3. Severity-wise Accident Count
4. Weather Condition Analysis
5. City-wise Accident Count
6. State-wise Average Severity
7. Day and Night Accident Analysis
8. Severity-wise Average Distance
9. Timezone-wise Accident Count
10. Weather and Severity Analysis

| Show                     | 25                                                                                  | entries    |            |            |      |               |             |            |                                             |                                                                                       | Search: |  |
|--------------------------|-------------------------------------------------------------------------------------|------------|------------|------------|------|---------------|-------------|------------|---------------------------------------------|---------------------------------------------------------------------------------------|---------|--|
| <input type="checkbox"/> |  | Permission | Owner      | Group      | Size | Last Modified | Replication | Block Size | Name                                        | <input type="checkbox"/>                                                              |         |  |
| <input type="checkbox"/> |                                                                                     | drwxr-xr-x | TechSource | supergroup | 0 B  | Sep 08 00:04  | 0           | 0 B        | <a href="#">task10_weather_severity</a>     |  |         |  |
| <input type="checkbox"/> |                                                                                     | drwxr-xr-x | TechSource | supergroup | 0 B  | Sep 07 17:13  | 0           | 0 B        | <a href="#">task1_severe</a>                |  |         |  |
| <input type="checkbox"/> |                                                                                     | drwxr-xr-x | TechSource | supergroup | 0 B  | Sep 07 18:32  | 0           | 0 B        | <a href="#">task2_state_count</a>           |  |         |  |
| <input type="checkbox"/> |                                                                                     | drwxr-xr-x | TechSource | supergroup | 0 B  | Sep 07 19:05  | 0           | 0 B        | <a href="#">task3_severity_count</a>        |  |         |  |
| <input type="checkbox"/> |                                                                                     | drwxr-xr-x | TechSource | supergroup | 0 B  | Sep 07 19:40  | 0           | 0 B        | <a href="#">task4_weather_count</a>         |  |         |  |
| <input type="checkbox"/> |                                                                                     | drwxr-xr-x | TechSource | supergroup | 0 B  | Sep 07 20:09  | 0           | 0 B        | <a href="#">task5_city_count</a>            |  |         |  |
| <input type="checkbox"/> |                                                                                     | drwxr-xr-x | TechSource | supergroup | 0 B  | Sep 07 21:41  | 0           | 0 B        | <a href="#">task6_state_avg_severity</a>    |  |         |  |
| <input type="checkbox"/> |                                                                                     | drwxr-xr-x | TechSource | supergroup | 0 B  | Sep 07 22:21  | 0           | 0 B        | <a href="#">task7_day_night</a>             |  |         |  |
| <input type="checkbox"/> |                                                                                     | drwxr-xr-x | TechSource | supergroup | 0 B  | Sep 07 22:52  | 0           | 0 B        | <a href="#">task8_severity_avg_distance</a> |  |         |  |
| <input type="checkbox"/> |                                                                                     | drwxr-xr-x | TechSource | supergroup | 0 B  | Sep 07 23:29  | 0           | 0 B        | <a href="#">task9_timezone_count</a>        |  |         |  |

### 8.1 Pig Task 1: Severe Accidents

Filters records where Severity equals 4 and stores the resulting severe-accident records in /usaccidents/pig/task1\_severe.

```

C:\BigDataProject\src>set HADOOP_USER_NAME=TechSource
C:\BigDataProject\src>echo %HADOOP_USER_NAME%
TechSource
C:\BigDataProject\src>hdfs dfs -ls -h /usaccidents/input/US_Accidents_1.5M.csv
-rw-r--r-- 1 Tech_Source supergroup 540.7 M 2026-09-08 23:28 /usaccidents/input/US_Accidents_1.5M.csv
C:\BigDataProject\src>pig PigTask1.pig
2026-09-07 17:12:13,752 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-07 17:12:13,753 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-09-07 17:12:13,755 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-09-07 17:12:13,138 [main] INFO org.apache.pig.Main - Apache Pig version 0.17.0 (r1797386) compiled Jun 02 2017, 15:41:58
2026-09-07 17:12:13,138 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig_1788779533127.1
2026-09-07 17:12:13,825 [main] INFO org.apache.pig.impl.util.Utils - Default bootup file C:\Users\Tech Source/.pigbootu
p not found
2026-09-07 17:12:13,924 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated
. Instead, use mapreduce.jobtracker.address
2026-09-07 17:12:13,925 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoo
op file system at: hdfs://localhost:9000
2026-09-07 17:12:14,439 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-PigTask1.pig-0aa7bbd2
-e2b5-4a8a-9684-ea515e5a1c85
2026-09-07 17:12:14,439 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled set
to false
2026-09-07 17:12:15,604 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.separator
is deprecated. Instead, use mapreduce.output.textoutputformat.separator
2026-09-07 17:12:15,633 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: FILTER

```

## Output for Severe Accidents

Block ID: 1073742199  
Block Pool ID: BP-79721839-192.168.0.183-1787044306402  
Generation Stamp: 1375  
Size: 72525  
Availability:  
• NASRIN

### File contents

GA,336555  
TX,164304  
FL,142758  
SC,83286  
PA,64569  
NY,62698  
IL,60311  
NC,51521

## 8.2 Pig Task 2: State-wise Count

Groups the dataset by State and calculates the number of accidents in each state. Output path: /usaccidents/pig/task2\_state\_count.

```

C:\BigDataProject\src>pig PigTask2.pig
2026-09-07 18:20:10,444 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-07 18:20:10,447 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-09-07 18:20:10,447 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-09-07 18:20:10,851 [main] INFO org.apache.pig.Main - Apache Pig version 0.17.0 (r1797386) compiled Jun 02 2017, 15:41:58
2026-09-07 18:20:10,851 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig_1788783610840.1
og
2026-09-07 18:20:11,470 [main] INFO org.apache.pig.impl.util.Utils - Default bootup file C:\Users\Tech Source/.pigbootu
p not found
2026-09-07 18:20:11,545 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated
. Instead, use mapreduce.jobtracker.address
2026-09-07 18:20:11,545 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoo
op file system at: hdfs://localhost:9000
2026-09-07 18:20:12,181 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-PigTask2.pig-10ec91d4
-d4ad-4426-b3a1-bff4930867f8
2026-09-07 18:20:12,182 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled set
to false
2026-09-07 18:20:13,352 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.separator
is deprecated. Instead, use mapreduce.output.textoutputformat.separator
2026-09-07 18:20:13,396 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: GROUP_
BY,ORDER_BY,FILTER
2026-09-07 18:20:13,446 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... will
not generate code.
2026-09-07 18:20:13,581 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENABLED=[Add
ForEach, ColumnMapKeyPrune, ConstantCalculator, GroupByConstParallelSetter, LimitOptimizer, LoadTypeCastInserter, MergeF
ilter, MergeForEach, NestedJoinOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer, PushDownForEachFlatten

```

## Output for State-wise Count

Block ID: 1073742238  
Block Pool ID: BP-79721839-192.168.0.183-1787044306402  
Generation Stamp: 1414  
Size: 413  
Availability:  
• NASRIN

### File contents

WY,3.62  
MT,3.6  
WV,3.136094674556213  
MD,2.7159017220280826  
CT,2.6878048780487807  
KS,2.6818181818181817  
ID,2.657142857142857  
IN,2.647880539499037

### 8.3 Pig Task 3: Severity-wise Count

Groups records by Severity and counts the records in each severity category. Output path: /usaccidents/pig/task3\_severity\_count.

```
C:\BigDataProject\src>notepad PigTask3.pig

C:\BigDataProject\src>hdfs dfs -rm -r /usaccidents/pig/task3_severity_count
rm: '/usaccidents/pig/task3_severity_count': No such file or directory

C:\BigDataProject\src>set HADOOP_USER_NAME=TechSource
HADOOP_USER_NAME=TechSource

C:\BigDataProject\src>echo %HADOOP_USER_NAME%
TechSource

C:\BigDataProject\src>pig PigTask3.pig
2026-09-07 18:52:46,713 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-07 18:52:46,715 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-09-07 18:52:46,715 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-09-07 18:52:47,122 [main] INFO org.apache.pig.Main - Apache Pig version 0.17.0 (r1797386) compiled Jun 02 2017, 15:41:58
2026-09-07 18:52:47,122 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig_1788785567114.1
2026-09-07 18:52:47,768 [main] INFO org.apache.pig.impl.util.Utils - Default bootstrap file C:\Users\Tech Source/.pigbootstrap not found
2026-09-07 18:52:47,840 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-09-07 18:52:47,840 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoop file system at: hdfs://localhost:9000
2026-09-07 18:52:48,492 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-PigTask3.pig-d02c2dbe
2026-09-07 18:52:48,492 [main] INFO org.apache.pig.PigServer - Connecting to hadoop file system at: hdfs://localhost:9000
```

### Output for Severity-wise Count

Block ID: 1073742277  
Block Pool ID: BP-79721839-192.168.0.183-1787044306402  
Generation Stamp: 1453  
Size: 33  
Availability:  
• NASRIN

#### File contents

```
r,False,False,False,False,False,False,False,False,False,False,False,False,False,Night,Night,Night,Night,Night
A-708949,Source2,4,2022-02-15 05:07:45,2022-02-15
06:41:56,31.53224,-81.445053,,15.229999542236328,Left hand shoulder blocked and left lane
blocked due to crash on I-95 Northbound between Exit 49 GA-251 and Exit 58 GA-57.,GA
Highway 57,Townsend,McIntosh,GA,31331,US,US/Eastern,KBQK,2022-02-15
05:15:00,34.0,34.0,88.0,30.42,10.0,CALM,0.0,0.0,Fair,False,False,False,False,False,False,False,False,False,False,False,False,False,Night,Night,Night,Night,Night
```

### 8.4 Pig Task 4: Weather Condition Count

Groups records by Weather\_Condition and counts the frequency of each weather condition. Output path: /usaccidents/pig/task4\_weather\_count.

```
C:\BigDataProject\src>set HADOOP_USER_NAME=TechSource
HADOOP_USER_NAME=TechSource

C:\BigDataProject\src>echo %HADOOP_USER_NAME%
TechSource

C:\BigDataProject\src>pig PigTask4.pig
2026-09-07 19:27:57,693 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-07 19:27:57,694 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-09-07 19:27:57,694 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-09-07 19:27:58,107 [main] INFO org.apache.pig.Main - Apache Pig version 0.17.0 (r1797386) compiled Jun 02 2017, 15:41:58
2026-09-07 19:27:58,107 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig_1788787678102.1
2026-09-07 19:27:58,715 [main] INFO org.apache.pig.impl.util.Utils - Default bootstrap file C:\Users\Tech Source/.pigbootstrap not found
2026-09-07 19:27:58,791 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-09-07 19:27:58,791 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoop file system at: hdfs://localhost:9000
2026-09-07 19:27:59,432 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-PigTask4.pig-00013dce-d612-4f3f-a9e6-a43fbd82fc29
2026-09-07 19:27:59,432 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled set to false
2026-09-07 19:28:00,480 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.separator is deprecated. Instead, use mapreduce.output.textoutputformat.separator
2026-09-07 19:28:00,516 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: GROUP_BY, ORDER_BY, FILTER
2026-09-07 19:28:00,553 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key Pair schemaTuple1 with key set ... will
```

## Output for Weather Condition Count

|                                                        |
|--------------------------------------------------------|
| Block ID: 1073742316                                   |
| Block Pool ID: BP-79721839-192.168.0.183-1787044306402 |
| Generation Stamp: 1492                                 |
| Size: 2298                                             |
| Availability:                                          |
| • NASRIN                                               |
| File contents                                          |
| Houston,53364                                          |
| Dallas,37903                                           |
| Los Angeles,33874                                      |
| Austin,27416                                           |
| Charlotte,26184                                        |
| Atlanta,19132                                          |
| Miami,16468                                            |
| Orlando,14969                                          |

## 8.5 Pig Task 5: City-wise Count

Groups records by City and counts accidents for each city. Output path: /usaccidents/pig/task5\_city\_count.

```
C:\BigDataProject\src>set HADOOP_USER_NAME=TechSource

C:\BigDataProject\src>echo %HADOOP_USER_NAME%
TechSource

C:\BigDataProject\src>pig PigTask5.pig
2026-09-07 19:58:09,462 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-07 19:58:09,462 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-09-07 19:58:09,464 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-09-07 19:58:09,666 [main] INFO org.apache.pig.Main - Apache Pig version 0.17.0 (r1797386) compiled Jun 02 2017, 15:41:58
2026-09-07 19:58:09,666 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig_1788789489661.10g
2026-09-07 19:58:10,018 [main] INFO org.apache.pig.impl.util.Utils - Default bootup file C:\Users\Tech Source/.pigbootup not found
2026-09-07 19:58:10,062 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated
```

## Output for City-wise Count

|                                                        |
|--------------------------------------------------------|
| Block ID: 1073742355                                   |
| Block Pool ID: BP-79721839-192.168.0.183-1787044306402 |
| Generation Stamp: 1531                                 |
| Size: 126152                                           |
| Availability:                                          |
| • NASRIN                                               |
| File contents                                          |
| CA,336555                                              |
| TX,164364                                              |
| FL,142758                                              |
| SC,83286                                               |
| PA,64569                                               |
| NY,62698                                               |
| IL,60311                                               |
| NC,51521                                               |

## 8.6 Pig Task 6: Average Severity by State

Groups records by State and calculates AVG(Severity), providing an average severity value for each state. Output path: /usaccidents/pig/task6\_state\_avg\_severity.

```

C:\BigDataProject\src>set HADOOP_USER_NAME=TechSource

C:\BigDataProject\src>echo %HADOOP_USER_NAME%
TechSource

C:\BigDataProject\src>pig PigTask6.pig
2026-09-07 21:29:03,077 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-07 21:29:03,078 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-09-07 21:29:03,078 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-09-07 21:29:03,277 [main] INFO org.apache.pig.Main - Apache Pig version 0.17.0 (r1797386) compiled Jun 02 2017, 15:41:58
2026-09-07 21:29:03,277 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig_1788794943272.1
og
2026-09-07 21:29:03,636 [main] INFO org.apache.pig.impl.util.Utils - Default bootup file C:\Users\Tech Source/.pigbootu
p not found
2026-09-07 21:29:03,678 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated
. Instead, use mapreduce.jobtracker.address
2026-09-07 21:29:03,678 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hado
op file system at: hdfs://localhost:9000
2026-09-07 21:29:04,049 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-PigTask6.pig-03accf94

```

## Output for Average Severity by State

|                                                                                                     |
|-----------------------------------------------------------------------------------------------------|
| Block ID: 1073742394                                                                                |
| Block Pool ID: BP-79721839-192.168.0.183-1787044306402                                              |
| Generation Stamp: 1570                                                                              |
| Size: 998                                                                                           |
| Availability: <ul style="list-style-type: none"> <li>NASRIN</li> </ul>                              |
| File contents                                                                                       |
| <pre> 1,0.0519272727275013149 2,0.09673681117568761 3,0.3528305848709385 4,4.801327574559954 </pre> |

## 8.7 Pig Task 7: Day/Night Accident Count

Uses Sunrise\_Sunset to group records into day/night categories and counts accidents for each category. Output path: /usaccidents/pig/task7\_day\_night.

```

C:\BigDataProject\src>pig PigTask7.pig
2026-09-07 22:10:09,646 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-07 22:10:09,647 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-09-07 22:10:09,647 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-09-07 22:10:09,841 [main] INFO org.apache.pig.Main - Apache Pig version 0.17.0 (r1797386) compiled Jun 02 2017, 15:41:58
2026-09-07 22:10:09,842 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig_1788797409837.1
og
2026-09-07 22:10:10,161 [main] INFO org.apache.pig.impl.util.Utils - Default bootup file C:\Users\Tech Source/.pigbootu
p not found
2026-09-07 22:10:10,200 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated
. Instead, use mapreduce.jobtracker.address
2026-09-07 22:10:10,200 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hado
op file system at: hdfs://localhost:9000
2026-09-07 22:10:10,573 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-PigTask7.pig-eb551a2c

```

## Output for Day/Night Accident Count

Block ID: 1073742433  
Block Pool ID: BP-79721839-192.168.0.183-1787044306402  
Generation Stamp: 1609  
Size: 42  
Availability:

- NASRIN

File contents

Fair,466102  
Clear,208905  
Mostly Cloudy,201302  
Cloudy,147317  
Partly Cloudy,136487  
Overcast,80265  
Light Rain,65383  
Scattered Clouds,47595

## 8.8 Pig Task 8: Average Distance by Severity

Groups records by Severity and calculates average Distance for each severity level. Output path: /usaccidents/pig/task8\_severity\_avg\_distance.

```
C:\BigDataProject\src>pig PigTask8.pig
2026-09-07 22:40:51,347 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-07 22:40:51,349 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-09-07 22:40:51,349 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-09-07 22:40:51,546 [main] INFO org.apache.pig.Main - Apache Pig version 0.17.0 (r1797386) compiled Jun 02 201
:41:58
2026-09-07 22:40:51,546 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig_1788799251
og
2026-09-07 22:40:51,867 [main] INFO org.apache.pig.impl.util.Utils - Default bootup file C:\Users\Tech Source\.pig
p not found
2026-09-07 22:40:51,905 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is depre
. Instead, use mapreduce.jobtracker.address
2026-09-07 22:40:51,906 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to
op file system at: hdfs://localhost:9000
2026-09-07 22:40:52,280 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-PigTask8.pig-3d5
-b136-42fe-92fe-65fb9a74717a
2026-09-07 22:40:52,281 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled
to false
2026-09-07 22:40:52,816 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.sep
r is deprecated. Instead, use mapreduce.output.textoutputformat.separator
2026-09-07 22:40:52,838 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: G
BY_ORDER_BY_FILTER
2026-09-07 22:40:52,861 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set...
not generate code.
```

## Output for Average Distance by Severity

Block ID: 1073742472  
Block Pool ID: BP-79721839-192.168.0.183-1787044306402  
Generation Stamp: 1648  
Size: 85  
Availability:

- NASRIN

File contents

ver Spring, Montgomery, MD, 20910, US, US/Eastern, KCGS, 2021-04-25  
07:15:00, 49.0, 48.0, 90.0, 29.68, 5.0, N, 6.0, 0.0, Cloudy, False, False, False, True, False, False, Fal  
se, False, False, False, False, Day, Day, Day, Day  
A-1009896, Source2, 4, 2021-04-25 17:20:22, 2021-04-25  
16:34:31, 39.420551, -77.484131, ..., 0.8309099737739563, One lane blocked due to accident on  
I-70 Eastbound from Exit 48 US-40 to Exit 49 US-40-ALT, I-70  
E, Frederick, Frederick, MD, 21703, US, US/Eastern, KFDK, 2021-04-25  
17:47:00, 64.0, 64.0, 45.0, 29.53, 10.0, NW, 15.0, 0.0, Mostly

## 8.9 Pig Task 9: Timezone-wise Count

Groups records by Timezone and counts accidents in each timezone category. Output path: /usaccidents/pig/task9\_timezone\_count.

```

C:\BigDataProject\src>pig PigTask9.pig
2026-09-07 23:16:19,253 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-07 23:16:19,254 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-09-07 23:16:19,255 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-09-07 23:16:19,710 [main] INFO org.apache.pig.Main - Apache Pig version 0.17.0 (r1797386) compiled Jun 02 2017, 15:41:58
2026-09-07 23:16:19,710 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig_1788801379702.1
2026-09-07 23:16:20,305 [main] INFO org.apache.pig.impl.util.Utils - Default bootup file C:\Users\Tech Source/.pigbootu
p not found
2026-09-07 23:16:20,370 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated
. Instead, use mapreduce.jobtracker.address
2026-09-07 23:16:20,370 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoo
op file system at: hdfs://localhost:9000
2026-09-07 23:16:20,983 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-PigTask9.pig-a5be0b1f
-ea85-44b2-aad9-79d3e45e74bc
2026-09-07 23:16:20,983 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled set
to false
2026-09-07 23:16:21,897 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.separato
r is deprecated. Instead, use mapreduce.output.textoutputformat.separator
2026-09-07 23:16:21,930 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: GROUP_
BY,ORDER BY,FILTER
2026-09-07 23:16:21,968 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... will
not generate code.
2026-09-07 23:16:22,016 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENABLED=[Add
ForEach, ColumnMapKeyPrune, ConstantCalculator, GroupByConstParallelSetter, LimitOptimizer, LoadTypeCastInserter, MergeF
ilter, MergeForEach, NestedLimitOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer, PushDownForEachFlatten,
PushUpFilter, SplitFilter, StreamTypeCastInserter]}

```

## Output for Timezone-wise Count

```

Block ID: 1073742511
Block Pool ID: BP-79721839-192.168.0.183-1787044306402
Generation Stamp: 1687
Size: 83
Availability:
  • NASRIN

File contents
Day,1092573
Night,405782
Sunrise_Sunset,1

```

## 8.10 Pig Task 10: Weather Condition and Severity

Groups records by the combination of Weather\_Condition and Severity and counts each combination. Output path: /usaccidents/pig/task10\_weather\_severity.

```

C:\BigDataProject\src>pig PigTask10.pig
2026-09-07 23:51:30,805 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-07 23:51:30,807 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-09-07 23:51:30,808 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-09-07 23:51:31,384 [main] INFO org.apache.pig.Main - Apache Pig version 0.17.0 (r1797386) compiled Jun 02 2017, 15:41:58
2026-09-07 23:51:31,384 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig_1788803491371.1
2026-09-07 23:51:32,265 [main] INFO org.apache.pig.impl.util.Utils - Default bootup file C:\Users\Tech Source/.pigbootu
p not found
2026-09-07 23:51:32,368 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated
. Instead, use mapreduce.jobtracker.address
2026-09-07 23:51:32,368 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoo
op file system at: hdfs://localhost:9000
2026-09-07 23:51:33,515 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-PigTask10.pig-9d85f3b
6-e5ff-40ed-a729-38e1164c38a7
2026-09-07 23:51:33,515 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled set
to false
2026-09-07 23:51:35,193 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.separato
r is deprecated. Instead, use mapreduce.output.textoutputformat.separator
2026-09-07 23:51:35,236 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: GROUP_
BY,ORDER BY,FILTER
2026-09-07 23:51:35,293 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... will
not generate code.
2026-09-07 23:51:35,364 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENABLED=[Add
ForEach, ColumnMapKeyPrune, ConstantCalculator, GroupByConstParallelSetter, LimitOptimizer, LoadTypeCastInserter, MergeF
ilter, MergeForEach, NestedLimitOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer, PushDownForEachFlatten,
PushUpFilter, SplitFilter, StreamTypeCastInserter]}

```

## Output for Weather Condition and Severity

|                                                        |
|--------------------------------------------------------|
| Block ID: 1073742550                                   |
| Block Pool ID: BP-79721839-192.168.0.183-1787044306402 |
| Generation Stamp: 1726                                 |
| Size: 6143                                             |
| Availability:                                          |
| • NASRIN                                               |
| File contents                                          |
| Day,1092573                                            |
| Night,405782                                           |
| Sunrise_Sunset,1                                       |

## 9. Results and Discussion

The project produced a collection of MapReduce and Pig outputs from the same HDFS input dataset. The results demonstrate that the dataset can be examined from multiple dimensions, including geography, severity, environment, distance, time of day, timezone, and traffic infrastructure. Because each analysis writes to a separate output directory, the generated results can be independently checked and used as evidence in the final report.

The verified state-wise MapReduce result shows substantial variation in accident counts among states. California has 336,555 recorded accident entries in the verified output, while Texas has 164,364 and Florida has 142,758. South Carolina has 83,286, Pennsylvania 64,569, and New York 62,698. This indicates that accident records are not uniformly distributed across the United States and that geographic aggregation can reveal strong differences between states.

The severity-wise analyses provide a different perspective. Rather than focusing on where accidents occur, severity grouping focuses on the seriousness category assigned to each record. The Pig and MapReduce approaches both provide frequency distributions that can be compared for consistency. Similarly, city-wise analysis can identify locations with relatively high numbers of recorded accident entries, while weather-condition analysis can summarize the environmental conditions associated with the records.

The average-severity-by-state Pig analysis extends simple counting by calculating an aggregate measure for each geographic group. The average-distance-by-severity analysis similarly connects accident seriousness with the distance field. These results should be interpreted as descriptive statistics of the dataset rather than as proof that one variable causes another. The day/night and timezone analyses add temporal and regional context, while the combined weather-condition-and-severity analysis allows two categorical dimensions to be examined together.

An important outcome of the project is the demonstration of consistency and complementary functionality between MapReduce and Pig. MapReduce requires explicit Java logic for reading fields, emitting key-value pairs, and reducing values. Pig can express many comparable operations using a smaller number of declarative data-flow statements. Therefore, MapReduce is useful when custom processing control is needed, while Pig is convenient for rapid exploratory analysis and repeated grouping/aggregation tasks.

For the final submitted report, screenshots of the actual command execution and output should be inserted under each task. Numeric values for tasks whose detailed output was not separately

recorded here should be copied directly from the corresponding HDFS part files rather than estimated.

## 10. Limitations and Future Scope

- The analysis is primarily descriptive and does not establish causal relationships between weather, road features, and accidents.
- CSV parsing with simple comma-delimited storage can be sensitive to commas inside text fields; a production pipeline would benefit from robust CSV parsing.
- Some fields contain missing values, so filtering and aggregation need careful handling.
- City-wise and weather-wise grouping can produce many unique categories, which may make direct manual inspection difficult.
- The project can be extended with Apache Hive for SQL-style queries and Apache Spark for faster in-memory distributed analytics.
- Future work could include visualization dashboards, time-series trend analysis, geospatial clustering, anomaly detection, and predictive modeling.
- Data-quality validation could be expanded to detect duplicate records, invalid coordinates, impossible durations, and inconsistent timestamps.

## 11. Conclusion

This project successfully demonstrates an end-to-end Big Data processing workflow using the US Accidents dataset and the Hadoop ecosystem. The work begins with environment preparation and verification of Java, Hadoop, and Apache Pig, followed by the organization and inspection of the input CSV. The dataset is then uploaded to Hadoop Distributed File System, where it becomes the common source for the subsequent distributed analyses. This sequence provides a clear example of how raw large-scale data can be moved from a local filesystem into a distributed processing environment before analytical computation begins.

The MapReduce component demonstrates the low-level programming model of Hadoop through five independent Java applications. State-wise accident counting provides geographic aggregation; severity-wise counting summarizes accident seriousness; city-wise counting provides a finer geographic view; weather-condition counting summarizes environmental categories; and traffic-signal counting analyzes a road-infrastructure indicator. Each program follows the Mapper and Reducer model and is packaged as a JAR for execution. The verified state-wise output, including 336,555 records for California, 164,364 for Texas, and 142,758 for Florida, demonstrates that the MapReduce pipeline successfully produced meaningful aggregate results from the HDFS dataset. The Apache Pig component extends the project beyond the five MapReduce operations. Ten Pig Latin scripts perform filtering, grouping, counting, averaging, and combined-field analysis. Severe accidents can be isolated through filtering, while state, severity, weather, and city counts provide multiple categorical summaries. Average severity by state and average distance by severity introduce numerical aggregation. Day/night and timezone analyses provide temporal and regional context, and the combined weather-condition-and-

severity analysis provides a two-dimensional view of accident records. These tasks demonstrate how Pig can reduce the amount of custom programming required for common Big Data transformations. The project also emphasizes verification and reproducibility. Separate HDFS output directories make it possible to inspect the result of each analytical task without overwriting another task's output. Command-line checks such as `hdfs dfs -ls` and `hdfs dfs -cat` allow the generated part files to be examined directly. In an academic laboratory setting, this is particularly useful because screenshots can show not only that a command was executed, but also that Hadoop successfully created and populated the expected output.

Overall, the project shows the complementary strengths of HDFS, MapReduce, and Apache Pig. HDFS provides the storage foundation, MapReduce provides detailed control over distributed computation, and Pig provides a higher-level data-flow approach for performing a wider variety of analyses efficiently. The project therefore connects Big Data theory with a practical implementation involving a real-world accident dataset. The resulting workflow can serve as a foundation for future work using Hive, Spark, visualization tools, statistical modeling, or machine-learning techniques to investigate road-safety patterns in greater depth.

## 12. References

1. **Hadoop Documentation** — Apache Hadoop, <https://hadoop.apache.org/>
2. **Apache Pig Documentation** — Apache Pig, <https://pig.apache.org/>
3. US Accidents dataset and accompanying dataset documentation used for this project.
4. **Project source files:** `StateCount.java`, `SeverityCount.java`, `CityCount.java`, `WeatherCount.java`, `TrafficSignalCount.java`.
5. **Project Pig scripts:** `PigTask1.pig` through `PigTask10.pig`.
6. <https://github.com/SiddhiPrabhu1995/YouTube-Data-Analysis-using-Hadoop/tree/master/PigAnalysis/scripts>